MIL-T-52530B
25 September 1980
SUPERSEDING
MIL-T-52530A
2 December 1975

MILITARY SPECIFICATION

TRAINING AID,

TAPE, MAGNETIC NORTH LINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tape with attached ring and counterweight used for training purposes.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-650	- Brazing Alloys, Copper, Copper-Zinc, and Copper-Phosphorus.
QQ-B-654	- Brazing Alloys, Silver.
QQ-S-781	- Strapping, Steel, and Seals.
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements for.
DDD-T-86	- Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6910

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for.

STANDARDS

FEDERAL

FED. Std. No. 751	- Stitches, Seams, and Stitchings.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 510 - General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel.
B 29 - Pig Lead.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The magnetic north line tape shall consist of a metal ring, cotton tape, and a lead counterweight. When assembled, the tape shall be in accordance with Figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Construction.

3.3.1 Ring. The ring shall be 11 gage (0.120 inch) steel conforming to ASTM A 510, UNS G10100-G10200. The outside diameter of the ring shall be 2.0 inches, plus or minus 0.062 inch. The junction point of the ring shall be welded in accordance with MIL-W-8611, or brazed in accordance with MIL-B-7883 using a silver-base filler metal conforming to QQ-B-654, Grade IV or QQ-B-650, Class BCuZnA or BCuZn-D. After welding or brazing, the ring shall be zinc plated in accordance with QQ-Z-325, Class 1, Type II.

3.3.2 Tape. The tape shall be cotton tape conforming to DDD-T-86, type optional, Class 3, dyed black, and shall be 0.50 inch wide and 102 inches long. The tape shall be stitched to the ring as shown on Figure 1. Stitching shall be in accordance with FED. STD. No. 751, Stitching Type EFb.

3.3.3 Counterweight. The counterweight shall be pig lead conforming to ASTM B 29, and shall be shaped as shown on Figure 1. A hole, 0.1875 inch in diameter, plus 0.062 inch minus 0 inch, shall be cast or drilled near the top (small end) of the counterweight. The counterweight shall be 2.75 inches long, plus or minus 0.125 inch. Maximum diameter of the counterweight shall be 1 inch. The counterweight shall weigh 6.0 ounces, plus or minus 1.0 ounce.

3.4 Identification marking. The magnetic north line tape shall be identified in accordance with MIL-STD-130.

3.5 Workmanship. The metal ring shall be free from sharp edges and burrs. The tape shall be free from frayed edges and stitching shall be clipped flush with the tape. The weight shall have a smooth surface and the hole shall be free from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The tape shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Dimensions not as specified.
103. Brazing or welding not as specified.
104. Plating not as specified.
105. Stitching not as specified.
106. Identification marking missing, incomplete or illegible.
107. Workmanship not as specified.

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, or containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.
109.	Consolidation not as specified for Level A.
110.	Gross weight or size exceeds the limitations of the box specification for Level A or Level B.
111.	Strapping not as specified for Level A.
112.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each tape, coiled to a minimum diameter, shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Consolidation. The preserved tapes shall be consolidated together in a close-fitting box conforming to PPP-B-636, W5c or W6c, style optional, in quantities not to exceed the weight limitation of the box specification.

5.1.2 Commercial. Each tape shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B or Commercial as specified (see 6.2).

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5.2.1 Level A. Tapes, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. The strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified herein, shall be Finish B. When specified (see 6.2), Finish A strapping shall be used.

5.2.2 Level B. Tapes, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional. The gross weight or size shall not exceed the limitation of the box specification.

5.2.3 Commercial. Tapes, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tape is a component of a map reading set and is used to illustrate magnetic north in teaching declination in basic map reading.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(c) When strapping shall be Finish A (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
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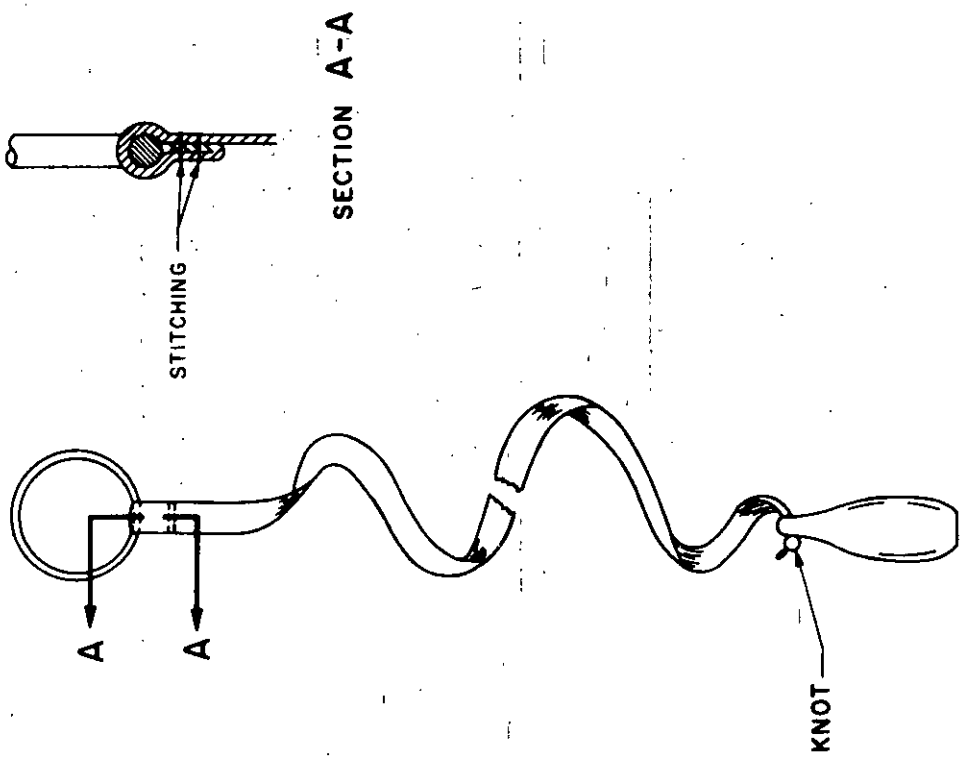


FIGURE 1. Tape, magnetic.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-52530B Training Aid, Tape, Magnetic North Line

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